

Fig. 1. Composite transistor with common well.

The transconductance g_{mb} varies from 20% to 30% of g_m for the same transistor in a CMOS process [2]. Therefore, it is possible to predict g_{mb} from g_m , and thus obtain the sizes of the transistors working in weak inversion.

An important configuration for MOS transistor in weak inversion is the composite transistor. The structure of an nMOS composite transistor is shown in Fig. 1 [17]. Consider that the transistors are implemented in the same well, since the substrate is common to all nMOS transistors in the CMOS process utilized [18].

The current and voltage expressions of the composite transistor can be derived directly from Fig. 1 and are given by (4), respectively. According to (1) and the conclusions given by [3], it is possible to relate current and voltage of the elements, as it will be explained

$$I_{DSa} = I_{DSb} \quad V_{DSa} = V_{GSa} - V_{GSb}. \quad (4)$$

Considering that the drain–source voltage applied is enough to saturate transistor Q_b , then V_{DSa} is given by (5). This expression is a function of the transistor sizes and the CMOS process parameters, valid for weak inversion only and is independent of the transistor's gate–source voltage [17]. Therefore, a variation in V_{DSb} does not affect V_{DSa} , so it is kept constant by cascode effect

$$V_{DSa} = \frac{kT}{q} \ln \left(1 + \frac{\left(\frac{W}{L}\right)_b}{\left(\frac{W}{L}\right)_a} \right). \quad (5)$$

Observe that the drain–source voltage for saturation of transistor Q_a does not depend on the gate–source voltage. This is the basis of the composite transistor, which is valid only for weak inversion operation and not for strong inversion.

III. IMPROVED MILLER OTA CIRCUIT

The circuit of an improved Miller operational transconductance amplifier (OTA) is shown in Fig. 2; thus called due to the improvements over the traditional Miller OTA [1]. A bulk-driven differential pair allows low-voltage operation of Miller OTA [2]. This is true because a transistor biased by the bulk and having maximum gate voltage is always active. In the circuit, if the common-mode voltage is close to the positive rail, the circuit is highly linear. On the other hand, if the common-mode voltage is close to the negative rail, the differential pair causes distortion to the signal, since the active load starts switching off. In order to solve this problem, two batteries are placed in series with the active load, to provide a dc shifting on the signal, as indicated in

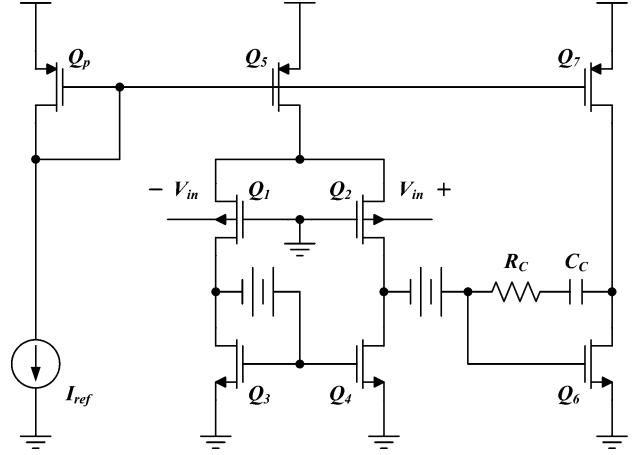


Fig. 2. Concept of the improved Miller OTA circuit.

Fig. 2. In this way, the active load remains operational with constant voltage for lower values of the input signal, thus avoiding non linearity conditions. If the common-mode voltage is close to the negative rail, the p–n junction of well–source of transistors Q_1 and Q_2 could be forward biased. Nevertheless, as it was experimentally confirmed by Blalock [11], it poses latch risk to the structure as long as the current in the bulk terminal is small, and should be carefully observed during the implementation of the project.

Obviously, circuits are not capable of generating voltages, but are capable of providing voltage drops, thus emulating batteries. A good option is the replacement of the battery by a common-gate amplifier. In terms of dc analysis, the transistor in diode configuration can be modeled by a resistance whose value is given by the inverse its transconductance, emulating a battery. Nevertheless, in terms of ac analysis, the transistor is an active load, and since the transistor is in saturation, it presents a gain. Therefore, it modifies the frequency response. The improved Miller OTA circuit is redrawn as shown in Fig. 3, in order to easily visualize the level shifters.

Observe that transistor pairs Q_{3a} – Q_{3b} and Q_{4a} – Q_{4b} form composite transistors. They allow the differential pair active load and the common gate amplifier to be biased by the same potential, without the use of additional biasing sources. From the composite transistor equations, it is possible to conclude that V_{DS3a} is given by (6). An analogous expression is obtained for V_{DS4a}

$$V_{DS3a} = \frac{kT}{q} \ln \left(1 + 2 \frac{\left(\frac{W}{L}\right)_{3b}}{\left(\frac{W}{L}\right)_{3a}} \right). \quad (6)$$

In terms of dc analysis, voltages V_{DS3a} and V_{DS4a} are equal and constant. Therefore, voltages V_{DS1} and V_{DS2} should be equal and constant, thus optimizing matching of differential pair Q_1 and Q_2 , consequently reducing the differential offset voltage. The dimensions of transistors Q_8 and Q_9 should be carefully matched in order to avoid unbalance in the structure of the differential pair. Additionally, care should be taken in dc bias between stages in order to avoid any systematic offset at the output of the differential amplifier. Therefore, from the expressions of currents as functions of the transistors sizes, it